

INSPACE-CAM-2026 Collision Avoidance Filing

Filing Reference ID:	3
Primary Spacecraft:	ISS (ZARYA) (NORAD 25544)
Secondary Spacecraft:	NORAD 900
Operator of Record:	ISRO Space Situational Awareness (SSA) Group
Time of Closest Approach (TCA):	2026-08-25 10:17:19.362061
Filing Status:	Filed
Submission Timestamp:	2026-08-25 10:19:14.328869

Regulatory Briefing & Maneuver Justification

INSPACE-CAM-2026 Collision Avoidance Filing Briefing

REGULATORY STATEMENT FOR SAT-TO-SAT CONJUNCTION MITIGATION

This official filing outlines the emergency collision avoidance maneuver (CAM) planned for ISS (ZARYA) (NORAD 25544) to mitigate a high-risk conjunction with CALSPHERE 1 (NORAD 900).

1. MANEUVER NECESSITY:

Based on high-fidelity orbital propagation and spatial screening, a close approach has been detected with a calculated Time of Closest Approach (TCA) at 2026-08-25T10:17:19.362061. The estimated miss distance is 0.35 km, yielding a Collision Probability (P_c) of 0.000145, which exceeds the standard safety threshold ($1.0e-4$). This maneuver is critical to safeguard space assets and eliminate collision risk.

2. TARGET SAFETY WINDOW:

The mitigation maneuver will execute approximately 12 hours prior to TCA, establishing a safe physical separation distance exceeding 2.0 km at the moment of closest approach. State vectors will be monitored continuously post-maneuver to verify trajectory stability.

3. IADC SPACE DEBRIS GUIDELINES COMPLIANCE:

All maneuvers are designed to minimize risk to the space environment. Post-maneuver trajectory propagation ensures that the satellite remains within a stable orbit, avoids secondary conjunctions with other cataloged objects, and fully complies with Inter-Agency Space Debris Coordination Committee (IADC) guidelines on space safety and mitigation.

Signed,
ISRO Space Situational Awareness (SSA) Group

Astrometric State Vectors & Covariance Matrices (TCA Epoch)

Below are the exact Earth-Centered Inertial (ECI) position (km) and velocity (km/s) state vectors and uncertainty covariances used to calculate the collision probability, ensuring absolute reproducibility for auditing

and regulatory verification.

Parameter	Primary Spacecraft	Secondary Spacecraft
ECI Position Vector (km)	[-2437.5742, 5562.6918, 3036.7529]	[1813.0903, 5857.7459, 4030.2898]
ECI Velocity Vector (km/s)	[-5.876648, 0.126754, -4.921875]	[-1.150871, -3.906411, 6.152978]
ECI Covariance Matrix (km^2)	[8.00e-02, 1.00e-02, 0.00e+00] [1.00e-02, 1.20e-01, 0.00e+00] [0.00e+00, 0.00e+00, 5.00e-02]	[5.00e-02, 0.00e+00, 0.00e+00] [0.00e+00, 8.00e-02, 0.00e+00] [0.00e+00, 0.00e+00, 3.00e-02]

Authorization & Sign-off

Authorized Operator Signature

Date